

FieldOps

AI Field Workforce Management Platform

System Design Document

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Apprend Technologies Pvt. Ltd.

Author	Apprend Technologies
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1. Executive Summary

FieldOps is an AI-powered field workforce management platform designed for Indian service businesses including HVAC, plumbing, electrical, pest control, and facility management companies. The platform combines GPS tracking, intelligent job scheduling, AI-optimized routing, digital proof-of-work, and automated invoicing into a single integrated solution.

India's field services market is estimated at \$25 billion and growing at 15% annually, driven by urbanization, commercial real estate expansion, and increasing expectations for professional service delivery. Yet most field service companies in India still manage operations through phone calls, WhatsApp groups, and paper-based job sheets, leading to inefficient routing, missed appointments, and poor customer communication.

FieldOps addresses these challenges with:

- AI-optimized job scheduling that minimizes travel time and maximizes jobs per worker per day
- Real-time GPS tracking with geofencing for attendance verification and job site arrival confirmation
- Skill-based job assignment matching the right technician to the right job
- Digital checklists and inspection forms that standardize service quality
- Photo and video proof-of-work capture for transparency and dispute resolution
- WhatsApp-based customer feedback collection leveraging India's preferred messaging platform
- Automated invoicing on job completion with digital payment collection
- Offline-first mobile app that works reliably in areas with poor connectivity

The platform aims to increase field worker productivity by 30% through optimized routing and scheduling, while reducing customer complaints by 50% through improved communication and transparency.

2. System Architecture

FieldOps uses a real-time, event-driven architecture designed to handle the unique challenges of field workforce management: intermittent connectivity, real-time location updates, and offline operation.

2.1 Technology Stack

Layer	Technology
Backend API	FastAPI (Python 3.12) with async WebSocket support
Web Dashboard	React / Next.js 14 with TypeScript
Mobile App	React Native with Expo for iOS and Android
Primary Database	PostgreSQL 16 with PostGIS for spatial queries
Cache & Queue	Redis 7 (BullMQ for job processing, pub/sub for real-time)
Real-time	WebSocket (Socket.IO) for live tracking dashboard
Maps & Routing	Google Maps Platform / OSRM for route optimization
Push Notifications	Firebase Cloud Messaging (FCM)
File Storage	S3-compatible (photos, videos, documents)
AI / ML	OpenRouter for AI scheduling; scikit-learn for demand prediction

2.2 Architecture Overview

The system consists of four primary components that communicate through events and real-time channels:

- **Admin Dashboard (Web):** Job management, worker management, scheduling, analytics, and customer management. Used by dispatchers and business owners.
- **Field Worker App (Mobile):** Job details, navigation, checklists, photo capture, time tracking, and status updates. Designed for use in the field with one-hand operation.
- **Real-Time Engine:** WebSocket server that pushes live location updates, job status changes, and notifications to the dashboard and mobile apps.
- **Background Workers:** BullMQ-powered workers for route optimization, invoice generation, feedback collection, and analytics computation.

2.3 Key Design Decisions

- **PostGIS spatial queries:** Nearest-worker lookup uses ST_DWithin and KNN-GiST indexes for sub-millisecond spatial queries even with thousands of workers.
- **Event sourcing for job state:** All job state transitions are stored as immutable events, enabling full audit trail and state reconstruction.
- **Offline-first mobile:** The mobile app uses WatermelonDB for local storage with background sync. Workers can complete jobs, fill checklists, and capture photos without connectivity.
- **Location batching:** GPS coordinates are collected every 30 seconds but uploaded in batches every 5 minutes to reduce battery and data usage.

3. Data Model

The database schema is designed for spatial operations and real-time tracking. All tables include `organization_id` for multi-tenant isolation. Location data uses PostGIS geometry types for efficient spatial indexing.

3.1 Core Tables

Table	Purpose	Key Fields
organizations	Tenant companies	name, industry, plan_tier, settings
workers	Field technicians	name, phone, skills[], team_id, status, last_location
teams	Worker groups/crews	name, leader_id, service_area (polygon)
jobs	Service jobs/work orders	title, type, priority, status, location (point), scheduled_at
job_assignments	Worker-job mappings	job_id, worker_id, assigned_at, started_at, completed_at
locations	GPS location history	worker_id, point (geometry), accuracy, battery_level, timestamp
routes	Optimized daily routes	worker_id, date, waypoints (JSONB), total_distance_km
time_entries	Attendance and time logs	worker_id, type (clock_in/out), location, verified
checklists	Inspection form templates	name, job_type, items (JSONB)
checklist_items	Completed checklist responses	job_id, checklist_id, responses (JSONB), completed_at
photos	Proof-of-work images	job_id, type (before/after), url, location, taken_at
customer_feedback	Post-service ratings	job_id, rating (1-5), comment, source (whatsapp/app)
invoices	Service invoices	job_id, amount_paise, status, payment_method
equipment	Tools and equipment registry	name, serial_no, assigned_to, last_served
inventory	Parts and supplies stock	name, sku, quantity, reorder_level, unit_cost_paise

3.2 Spatial Indexes

The following PostGIS indexes enable fast spatial queries critical for real-time operations:

- `workers.last_location`: GiST index for nearest-worker queries during emergency dispatch
- `jobs.location`: GiST index for geographic job clustering and zone-based assignment

- teams.service_area: GiST index for determining which team covers a given location
- locations (worker_id, timestamp): B-tree + GiST compound index for historical track playback

4. Core Features

4.1 AI-Optimized Job Scheduling and Routing

The scheduling engine uses a combination of constraint satisfaction and optimization algorithms to create optimal daily routes for each worker:

- Input: Unassigned jobs with location, time window, required skills, estimated duration, and priority
- Constraints: Worker availability, skills, maximum working hours, break requirements, and service area
- Optimization: Minimize total travel distance while respecting all constraints and time windows
- Output: Per-worker route with ordered job list, estimated arrival times, and turn-by-turn navigation
- Re-optimization: Routes are recalculated in real-time when new urgent jobs are added or workers become unavailable

The routing engine uses OSRM (Open Source Routing Machine) for distance/time matrix computation with Indian road network data, falling back to Google Maps Directions API for areas with poor OSM coverage.

4.2 Real-Time GPS Tracking

The platform provides real-time visibility into field worker locations:

- Live map view on admin dashboard showing all active workers with status indicators
- Geofencing: automatic clock-in when worker enters a job site, alerts when worker leaves designated area
- Historical track playback for reviewing daily routes and verifying attendance
- Battery-aware tracking: reduces GPS frequency when battery is low to prevent worker phones from dying
- Privacy controls: tracking pauses outside working hours; workers can see their own tracking data

4.3 Skill-Based Job Assignment

Jobs are matched to workers based on:

Factor	Description
Skills	Worker certifications and capabilities (e.g., AC repair, electrical, plumbing)
Proximity	Distance from current location to job site
Availability	Current job status, remaining capacity, and scheduled breaks
Performance	Historical completion rate, customer rating, and on-time record
Equipment	Required tools/parts currently in worker's possession

4.4 Digital Checklists and Inspection Forms

Standardized service checklists ensure consistent quality across all technicians:

- Configurable checklist templates per job type (AC service checklist differs from plumbing)
- Mandatory photo capture at specific checklist steps (e.g., before/after photos)
- Pass/fail items with mandatory notes on failure
- Conditional logic: follow-up questions based on previous answers
- Offline completion with sync when connectivity returns

4.5 Photo/Video Proof-of-Work

Workers capture timestamped, geotagged photos and videos as proof of work:

- Before and after photos for visual comparison of work performed
- Automatic metadata: GPS location, timestamp, and worker ID embedded in every capture
- Tamper detection: photos must be taken in-app (no gallery uploads) to prevent fraud
- Customer-facing photo gallery: customers can view proof-of-work photos in their portal
- Compressed upload: images are compressed on-device before upload to minimize data usage

4.6 Customer Communication

FieldOps keeps customers informed throughout the service lifecycle via WhatsApp:

- Appointment confirmation with technician name, photo, and expected arrival time
- Real-time ETA updates as the technician approaches (shared live location)
- Job completion notification with summary and before/after photos
- WhatsApp-based feedback collection (1-5 star rating with optional comment)
- Automated follow-up for recurring service reminders (e.g., annual AC service)

4.7 Additional Features

- Automated invoicing on job completion with digital payment collection (UPI, card)
- Equipment and inventory tracking with low-stock alerts and usage per job
- Worker performance analytics: jobs/day, avg completion time, customer rating, on-time %
- Customer portal for job status tracking, history, and upcoming appointments
- Emergency dispatch: one-click assignment to nearest available qualified worker
- Attendance and time tracking with geo-verification (must be at office/job site to clock in)

5. Mobile App Architecture

The FieldOps mobile app is the primary interface for field workers. It must be reliable, fast, and functional in areas with poor or no connectivity.

5.1 Offline-First Design

- Local database: WatermelonDB (SQLite-based) stores all job data, checklists, and pending uploads locally
- Background sync: Changes are queued and synced when connectivity is available, using a CRDT-based conflict resolution strategy
- Photo queue: Captured photos are stored locally and uploaded in the background with retry logic
- Graceful degradation: Core functionality (view jobs, complete checklists, capture photos) works fully offline

5.2 Battery Optimization

Field workers spend 8-10 hours in the field. Battery optimization is critical:

- Adaptive GPS: High frequency (30s) during transit, low frequency (5min) when stationary at job site
- Batch uploads: Network requests are batched to minimize radio wake-ups
- Background location: Uses the most battery-efficient location provider available (fused location on Android)
- Battery monitoring: Reduces tracking frequency below 20% battery; alerts dispatcher below 10%

5.3 Key Mobile Screens

Screen	Features
Today's Route	Ordered job list with map view, ETA, and navigation. Swipe to start/complete jobs.
Job Detail	Customer info, job description, checklist, photos, notes, and status controls.
Checklist	Step-by-step inspection form with photo capture and pass/fail items.
Camera	In-app camera with auto-geotagging, timestamp overlay, and compression.
Attendance	Clock in/out with geofence verification. Shows daily hours and overtime.
Inventory	Parts used per job, request parts, scan barcode to log usage.

6. API Design

FieldOps provides both REST and WebSocket APIs. REST for CRUD operations; WebSocket for real-time location tracking and status updates.

6.1 REST Endpoints

Method	Endpoint	Description
POST	/jobs	Create a new job/work order
GET	/jobs/unassigned	List unassigned jobs for scheduling
POST	/jobs/:id/assign	Assign worker to job
POST	/jobs/:id/complete	Mark job as completed
POST	/routes/optimize	Generate optimized daily routes
GET	/workers/:id/location	Get worker's current location
POST	/workers/nearest	Find nearest qualified workers to a location
POST	/feedback/collect	Trigger WhatsApp feedback collection

6.2 WebSocket Events

Event	Direction	Description
location:update	Client -> Server	Worker sends GPS coordinates batch
location:broadcast	Server -> Client	Dashboard receives live worker positions
job:status_changed	Server -> Client	Job status update (assigned, started, completed)
job:new_assignment	Server -> Client	New job assigned to worker (push to mobile)
worker:sos	Client -> Server	Worker emergency alert with location

7. Revenue Model

FieldOps uses a per-user pricing model that scales with team size:

Plan	Price	Features
Starter	Rs. 499/user/mo	Job management, GPS tracking, basic scheduling, up to 10 workers
Professional	Rs. 799/user/mo	AI routing, checklists, photo proof, customer portal, up to 50 workers
Enterprise	Rs. 1,199/user/mo	Emergency dispatch, inventory, analytics, API, unlimited workers
Admin Dashboard	Rs. 1,999/mo	Per-organization dashboard fee (included with 10+ worker seats)

Additional revenue from WhatsApp message fees (pass-through at cost + 10% margin) and premium integrations (Tally, SAP).

8. Integration

FieldOps integrates with internal products and third-party services:

8.1 Internal Integrations

- GoSumo: CRM integration for customer data sync; service history feeds into client management
- InvoiceAI: Completed jobs automatically generate GST-compliant invoices with AR automation

8.2 External Integrations

- Google Maps Platform: Maps display, geocoding, directions, and distance matrix for route optimization
- WhatsApp Business API: Customer notifications, appointment confirmations, feedback collection, and ETA sharing
- Firebase Cloud Messaging: Push notifications for new job assignments, schedule changes, and alerts
- Razorpay / UPI: Digital payment collection upon job completion
- Tally: Accounting data export for invoices and expense tracking
- Google Calendar: Job schedule sync for workers and customers

9. Security

FieldOps handles sensitive location and customer data requiring careful privacy and security controls:

- Location data privacy: GPS data is encrypted at rest and in transit; retained for a configurable period (default 90 days)
- Worker consent: Explicit opt-in for location tracking with clear disclosure of tracking hours and data usage

- Data minimization: Location precision is reduced for stored data (neighborhood level) while real-time data uses full precision
- Customer data protection: Customer contact information is visible only to assigned workers and dispatchers
- Role-based access: Organization Admin, Dispatcher, Team Lead, and Worker roles with graduated permissions
- Device security: App requires PIN/biometric to access; remote wipe capability for lost devices
- API security: JWT authentication with refresh tokens; API rate limiting; webhook signature verification
- DPDP Act compliance: Data processing agreements, consent management, and data subject request handling

10. Milestones

FieldOps follows a 14-week timeline with web dashboard first, mobile app second:

Week	Phase	Deliverables
1-3	Foundation	Database schema with PostGIS, job CRUD API, basic worker management, admin dashboard
4-6	MVP (Web)	Job scheduling, manual assignment, GPS tracking (web upload), map dashboard, basic routing
7-8	Mobile App v1	React Native app with job view, status updates, GPS tracking, photo capture, offline storage
9-10	Mobile + Web	Digital checklists, AI route optimization, real-time WebSocket tracking, customer notifications
11-12	Beta	Customer portal, automated invoicing, WhatsApp feedback, emergency dispatch, beta testing
13-14	GA Launch	Performance analytics, inventory tracking, production hardening, documentation, launch

Post-GA roadmap includes fleet vehicle tracking, predictive maintenance scheduling, multi-language support (Hindi, Kannada, Tamil), and integration with government skill certification databases (NSDC).